S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

AI/NLP Risk Engine + Tactical Index Rebalancer + Portfolio Stress Testing

Financial NLP, tactical portfolio rebalancing and scenario stress testing in one explainable decision-support prototype

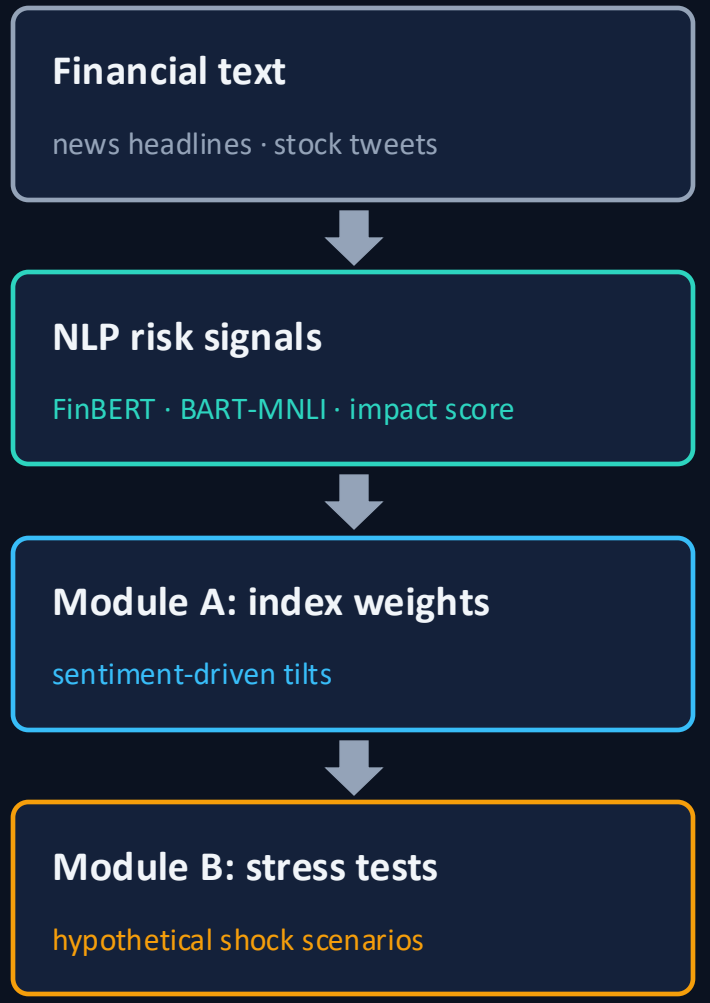
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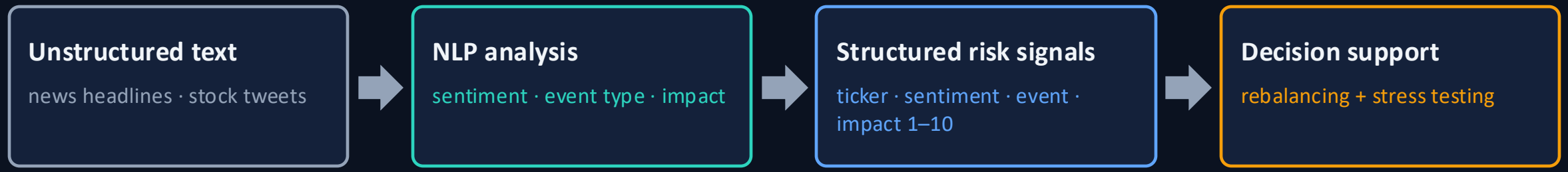
Demo Video Link: https://youtu.be/91JTRFemLJA

Slide Deck Link: Not hosted externally



From unstructured text to explainable signals

Financial news and social commentary carry risk information, but turning noisy, unstructured text into consistent, explainable portfolio signals is hard.



What the prototype demonstrates

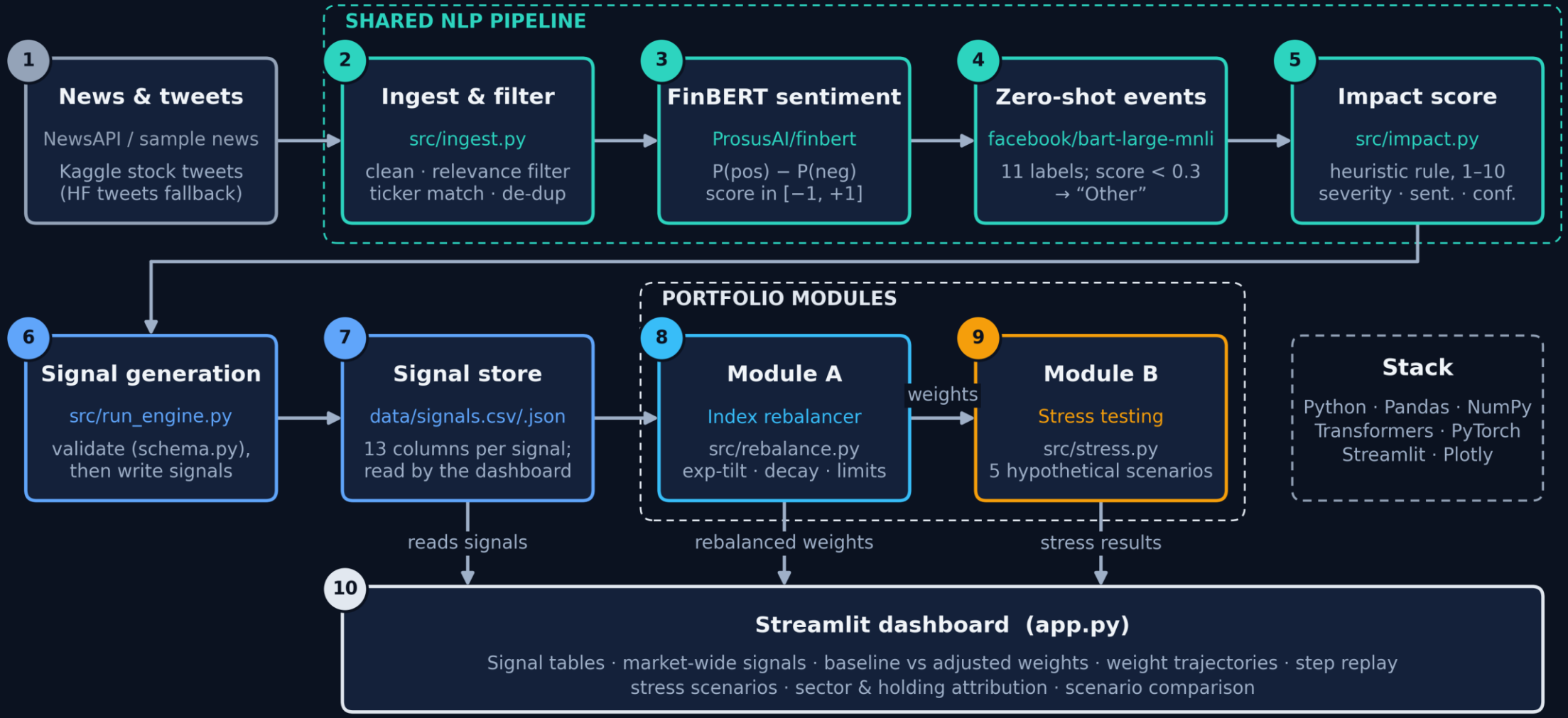
- Text becomes structured, searchable signals
- Signals tilt a mock index by transparent rules
- Resulting weights face hypothetical shocks

Why two modules complement each other

- A asks: how should news shift weights?
- B asks: how fragile is the resulting portfolio?
- B stress-tests A's final rebalanced weights

A prototype for analyst decision support, not an autonomous trading system

Architecture: one NLP pipeline, two modules



Engineering highlights: signals to weights

1. **Sentiment + event, together**
   FinBERT direction and BART-MNLI category in one record
2. **Transparent impact score**
   Hand-set severity × sentiment × confidence; inspectable
3. **Confidence-aware tickers**
   Exchange tag > cashtag > name + finance; multi-ticker kept
4. **Clean inputs**
   Relevance filter and >85% near-duplicate drop cut noise
5. **One expansion, two views**
   Replay display and rebalancer share signal expansion
6. **Auto time buckets, tunable knobs**
   Day to year closest to 20 steps; sensitivity, decay, caps
7. **Validated and tested**
   Schema check before write; 87 offline unit tests pass

Signal → weight (Module A)

IMPACT, 1–10

$$\text{severity} \times (0.5 + 0.5 \cdot |\text{sent}|) \times (0.5 + 0.5 \cdot \text{conf})$$

NET SIGNAL, PER STOCK/STEP

$$\Sigma\ \text{sentiment} \times \text{impact} / 10$$
clipped to ±2

TILT

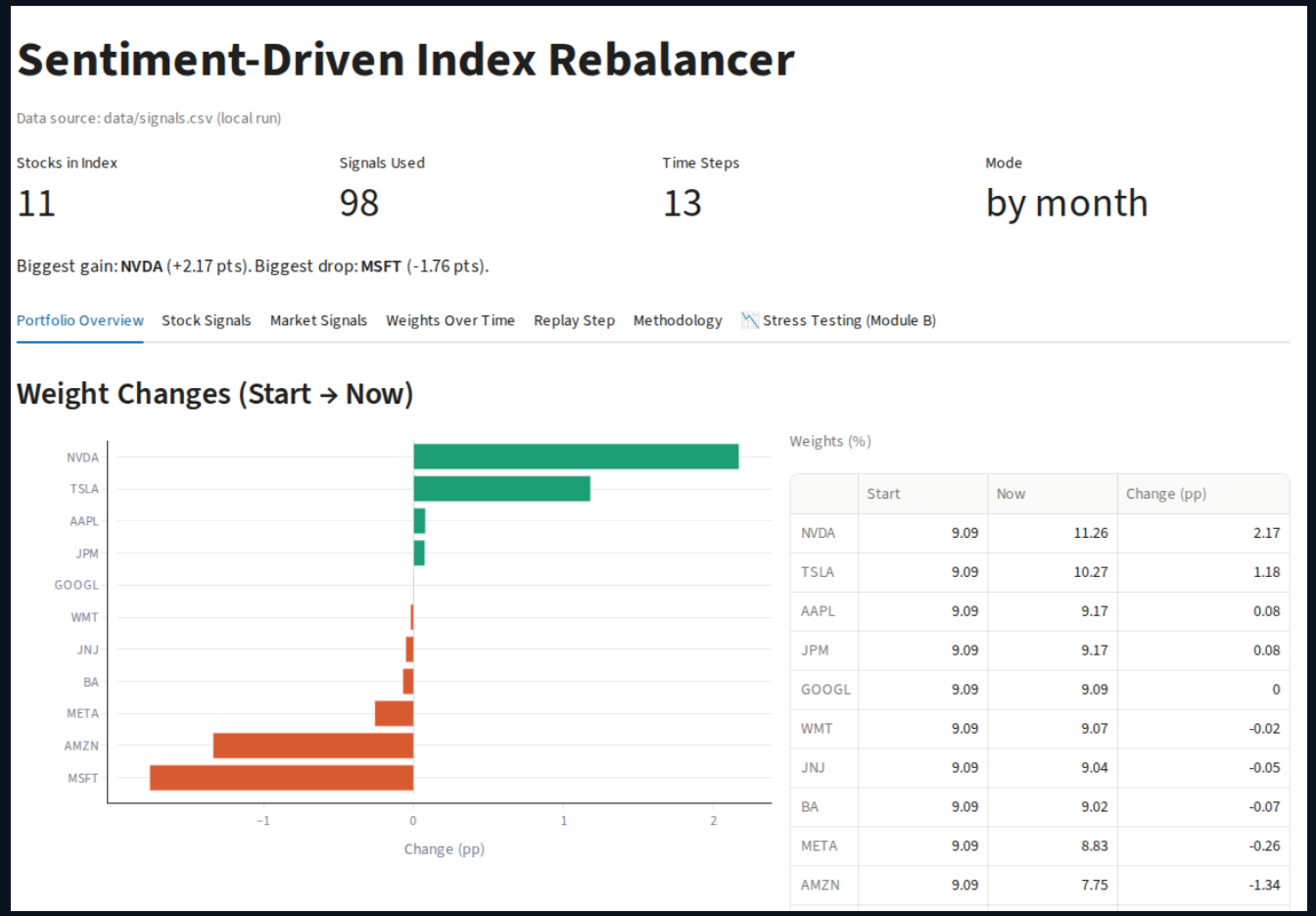
$$w \leftarrow w \cdot \exp(1.5 \times \text{net signal})$$
then normalise to 100%

DECAY + LIMITS

$$0.95 \cdot \text{tilt} + 0.05 \cdot w_0$$
clip each weight to 1%–15%

Defaults shown (sensitivity 1.5, decay 0.05); sliders adjust them.

Results: verifiable prototype outputs



Streamlit dashboard, Portfolio Overview. Committed data/signals.csv, default parameters. Weight changes are simulated tilts, not returns.

70% — **FinBERT accuracy, 0.63 macro-F1**
off-the-shelf model, 300 validation tweets

129 — **signals in the committed dataset**
76 tweets (Oct 2021–Sep 2022) + 53 headlines (9 Oct 2026)

13 — **monthly replay steps**
98 signals move 11 stocks: NVDA +2.17 pp, MSFT −1.76 pp

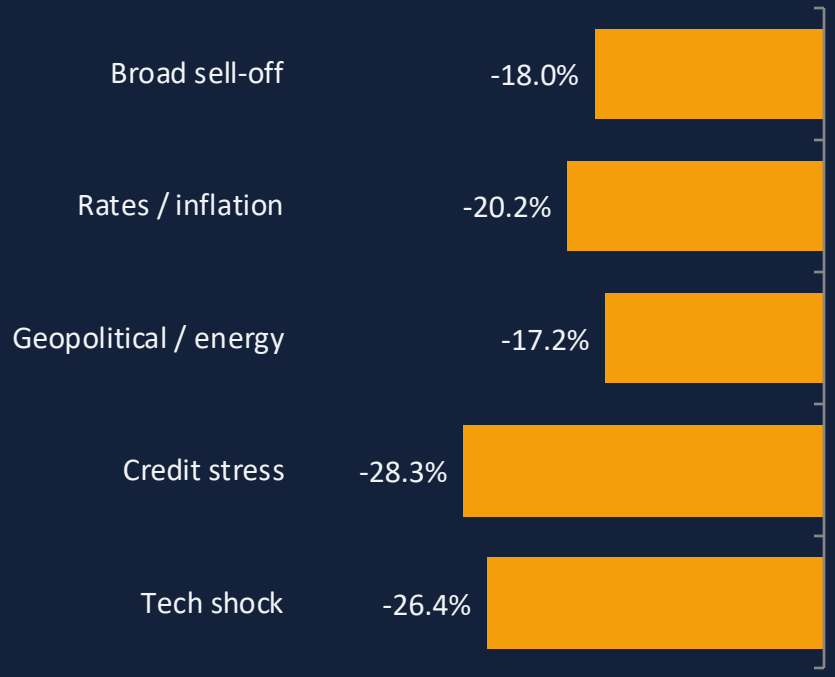
87 — **unit tests pass offline**
5 more tests need network access to load data

Domain impact: explainable decision support

1. **Structured, searchable signals**
   Text becomes ticker, sentiment, event and impact fields
2. **Explainable by design**
   Analysts trace each weight move to its source signals
3. **News-aware mock index**
   Rules-based tilts show how sentiment could shift weights
4. **Shock sensitivity testing**
   Hypothetical scenarios run on the rebalanced portfolio
5. **Where the risk sits**
   Scenario comparison, sector and holding attribution

Before real deployment
Stronger validation, reliable data feeds, risk controls, market-data calibration

Simulated portfolio impact by scenario



Hypothetical sector shocks applied to Module A's final weights (11 stocks). Not historical losses, forecasts or calibrated estimates.

Engineering boundaries and roadmap

Engineering boundary	Next step
Sentiment checked on 300 tweets; event labels not scored against ground truth	Larger labeled evaluation for sentiment and event classes
Impact weights are hand-chosen heuristics	Calibrate impact to observed market reactions
Ticker matching is keyword-based and can misattribute	Stronger entity resolution and ticker attribution
Module A has no historical price backtest	Market-data integration and walk-forward backtesting
Module B shocks are hypothetical, with no volatility or correlation calibration	Calibrate risk parameters on historical market data
Sparse, gapped data: tweets Oct 2021–Sep 2022, then 53 headlines from one 45-minute window in Oct 2026	Continuous multi-source feed, data-quality monitoring, reproducible evaluation

TAKEAWAY

A working prototype that connects financial NLP to explainable portfolio decision-support workflows.